nature genetics

Article

https://doi.org/10.1038/s41588-025-02441-6

Designing synthetic regulatory elements using the generative AI framework DNA-Diffusion

Received: 21 June 2024

Accepted: 1 November 2025

Published online: 23 December 2025

Check for updates

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Systematically designing regulatory elements for precise gene expression control remains a central challenge in genomics and synthetic biology. Here we introduce DNA-Diffusion, a generative artificial intelligence framework that uses machine learning trained on DNA accessibility data from diverse cell lines to design compact regulatory elements with cell-type-specific activity. We show that DNA-Diffusion generates 200-base-pair synthetic elements that recapitulate endogenous transcription factor binding grammar while exhibiting enhanced cell-type specificity. We validated these elements using a 5,850-element STARR-seq library across three cell lines. Moreover, we demonstrated successful endogenous gene modulation using EXTRA-seq, reactivating *AXIN2*, a leukemia-protective gene, in its native genomic context. Our approach outperforms existing computational methods in balancing functional activity with cell-type specificity while maintaining sequence diversity. This work establishes DNA-Diffusion as a powerful tool for engineering compact, highly specific regulatory elements crucial for advancing gene therapies and understanding gene regulation.

Gene regulation is a complex process in which elements such as promoters and enhancers modulate gene activity, defining cell identity. A key goal in genomics and synthetic biology is to harness these elements for gene expression control and to correct disease-related misregulation. Although large-scale projects[1–7] have annotated millions of regulatory elements, synthesizing them de novo to achieve desired gene expression remains a critical challenge. Generative artificial intelligence (AI) has revolutionized various fields and offers a powerful approach to design sequences, enabling precise control (Fig. 1a,b and Supplementary Note 1).

Diffusion models[8–10], a class of generative AI, have shown success in generating complex data such as images and proteins[11–14], making them a promising tool for designing biological sequences (Supplementary Note 2). However, current AI methods for designing regulatory elements[15–23] face important limitations. They often lack user control over activity levels, perform inconsistently across different cell types and, most critically, have not been validated in their native genomic context, relying instead on artificial episomal assays (Supplementary Note 3).

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To overcome these challenges, we introduce DNA-Diffusion, a conditional diffusion model for designing context-specific DNA regulatory sequences. Its key innovations include: (1) a single model that generates cell-type-specific elements without requiring separate models for each cell type or additional postprocessing steps; (2) tunable control to balance signal strength and sequence diversity; and (3) the ability to generate sequences that preserve natural regulatory grammar while exhibiting enhanced and specific activity. This approach is vital for applications such as targeted gene therapies[24], where compact and highly specific elements are necessary for delivery systems such as adeno-associated viruses (AAVs), which have limited cargo space[25]. Given the limitation of using endogenous enhancers in therapeutic applications[26], DNA-Diffusion generates tailored, compact 200-bp elements, expanding the toolkit for precision medicine (Supplementary Note 4).

We rigorously validated DNA-Diffusion through a multifaceted approach. Extensive in silico analyses confirmed the sequences had appropriate transcription factor (TF) binding motifs and regulatory patterns. We also experimentally assessed the transcriptional activity of 5,850 synthetic sequences using large-scale self-transcribing active regulatory region sequencing (STARR-seq)[27,28]. Critically, we validated their regulatory function in native genomic context using an extended reporter assay followed by sequencing (EXTRA-seq)[29], successfully modulating *AXIN2*, a leukemia therapeutic target[30], at its native locus in human B cells. Finally, comprehensive benchmarking against existing methods revealed DNA-Diffusion's superior signal intensity, specificity and sequence diversity balance. DNA-Diffusion is a diffusion model for regulatory element design validated through both large-scale reporter assays and direct endogenous gene modulation, bridging the gap between computational design and therapeutic application.

Results

DNA-Diffusion is a conditional diffusion model that generates cell-type-specific regulatory elements

DNA-Diffusion is a conditional diffusion model that operates in the space of DNA sequences. Sequences are encoded using a strategy akin to one-hot encoding, but each nucleotide has a support range of −1 to 1 to facilitate the injection of Gaussian noise centered around zero. The model's backbone is a denoising U-Net[31] architecture with two embedding layers for cell-type labels and diffusion timesteps, respectively, enabling it to learn a cell-type-specific denoising function (Fig. 1c; Methods). The model receives three inputs during training: DNA sequences, a timestep and a cell-type label. After training, a cell-type label can be used as input to generate novel, cell-type-specific sequences. To train the model, we leverage the comprehensive DNase I hypersensitive site (DHS) index dataset curated by Meuleman et al.[32], which contains 733 biosamples from 438 cell and tissue types. We focused our initial efforts on three well-characterized and distinct cell lines: GM12878 (B-lymphocyte), K562 (leukemia) and HepG2 (hepatocellular carcinoma). These lines were chosen for their diverse biological origins, distinct germ layer lineages, ease of culture and the rich availability of existing genomic data for model interpretability and evaluation.

We assigned a categorical class label for each cell type and selected sequences spanning 200 bp around the summit of cell-type-specific DHS peaks (Methods). To ensure model generalizability and prevent data leakage, we partitioned the dataset using a chromosome-based stratified sampling strategy: chromosome 1 was held out as the independent test set, chromosome 2 was used for validation, and the remaining chromosomes were used for training. During training, the model learns to reverse a forward process where varying levels of Gaussian noise are progressively added to the encoded sequence representations. The model's objective is to learn a function that can effectively denoise a sequence at each step by predicting the patterns of introduced noise (Fig. 1d). After successful training (Supplementary Fig. 1), the model can synthesize novel DNA sequences from random noise through a reverse process, conditioned on a specific cell-type label and a number of steps (Fig. 1e). With this model, we generated 100,000 synthetic 200-bp DNA sequences for each of the three cell types and evaluated their genomic and functional characteristic across various metrics: sequence uniqueness, TF motif composition, predicted accessibility and the predicted capability to drive gene expression in a cell-type-specific manner (Fig. 1f,g).

Diffusion-generated sequences exhibit motifs present in endogenous sequences without memorizing existing sequences

To verify the uniqueness of the DNA-Diffusion sequences and confirm that they were not mere copies of the training sequences, we used BLAT[33] to conduct a comparative sequence alignment between the generated DNA-Diffusion sequences and the sets of training, test and random genomic coordinates.

DNA-Diffusion sequences showed minimal sequence identity to the training set, with only 4.7% having at least one alignment hit of ≥20 bp, substantially lower than that of randomly selected genomic coordinates (23.1%) and test DHSs (14.5%), indicating genuinely novel DNA sequences without overfitting (Fig. 2a, top). The within-group alignment for generated sequences (25.3%) was higher than for the training sequences (14.5%), which is expected as the model learns to combine common functional TF motifs (Fig. 2a, bottom).

After accounting for repetitive elements that inflated random sequence alignments (Supplementary Fig. 2a,b), and using a more stringent ≥30-bp threshold, only 0.6% of DNA-Diffusion sequences aligned to the training set (Supplementary Fig. 3a and Supplementary Note 5). These alignment results demonstrate that DNA-Diffusion generates diverse sequences without memorizing training data.

Fig. 1 | Overview of DNA-Diffusion framework for generative modeling, validation, and interpretation of cell-type-specific regulatory sequences. **a**, Examples of generative AI models applied to language, image generation and biological problems. **b**, Examples of potential therapeutic applications based on synthetic cell-type-specific sequences. **c**, A schematic overview of the DNA-Diffusion model. The model utilizes a U-Net architecture to iteratively generate new DNA sequences based on cell types presented in the training dataset. **d**, The DNA-Diffusion model was trained using unique DNase I hypersensitive sites (DNase I) DNA sequences from different cell types, including K562, HepG2 and GM12878. The training involves transforming the endogenous DHS sequence into a hot-encoded format and introducing a fixed amount of standard normal noise. The trained U-Net uses the expected noise level (determined by the timestep) and cell-type information to predict and remove the added noise. This noise prediction process is repeated during training across the entire sequence dataset with varied noise intensities. **e**, To generate a new sequence given a cell-type label, a hot-encoded DNA matrix with random Gaussian noise is generated, and the U-Net iteratively denoises this matrix over 50 steps, progressively converging to a sequence that reflects the characteristics of the target cell type. **f**, Different in silico validations were utilized to evaluate the accessibility, regulatory activity and motif composition of DNA-Diffusion and endogenous DHS regions. **g**, A framework was developed to select and interpret generated sequences based on cell-type signal specificity, intensity or motif composition. **h**, A schematic depicting experimental protocols for DNA-Diffusion sequence validation. Left: an episomal STARR-seq experiment that examines enhancer activity using self-reporter plasmids. Right: endogenous integration experiment that measures expression of target gene *AXIN2*. **i**, A schematic depicting sequence interpretation metrics to assist with analyzing how motif patterns drive expression in a cell-type-specific manner. **j**, Benchmarking framework for analyzing current state-of-the-art deep learning-based sequence design models. Signal intensity, signal specificity and motif diversity are the metrics used to evaluate sequences from these models.

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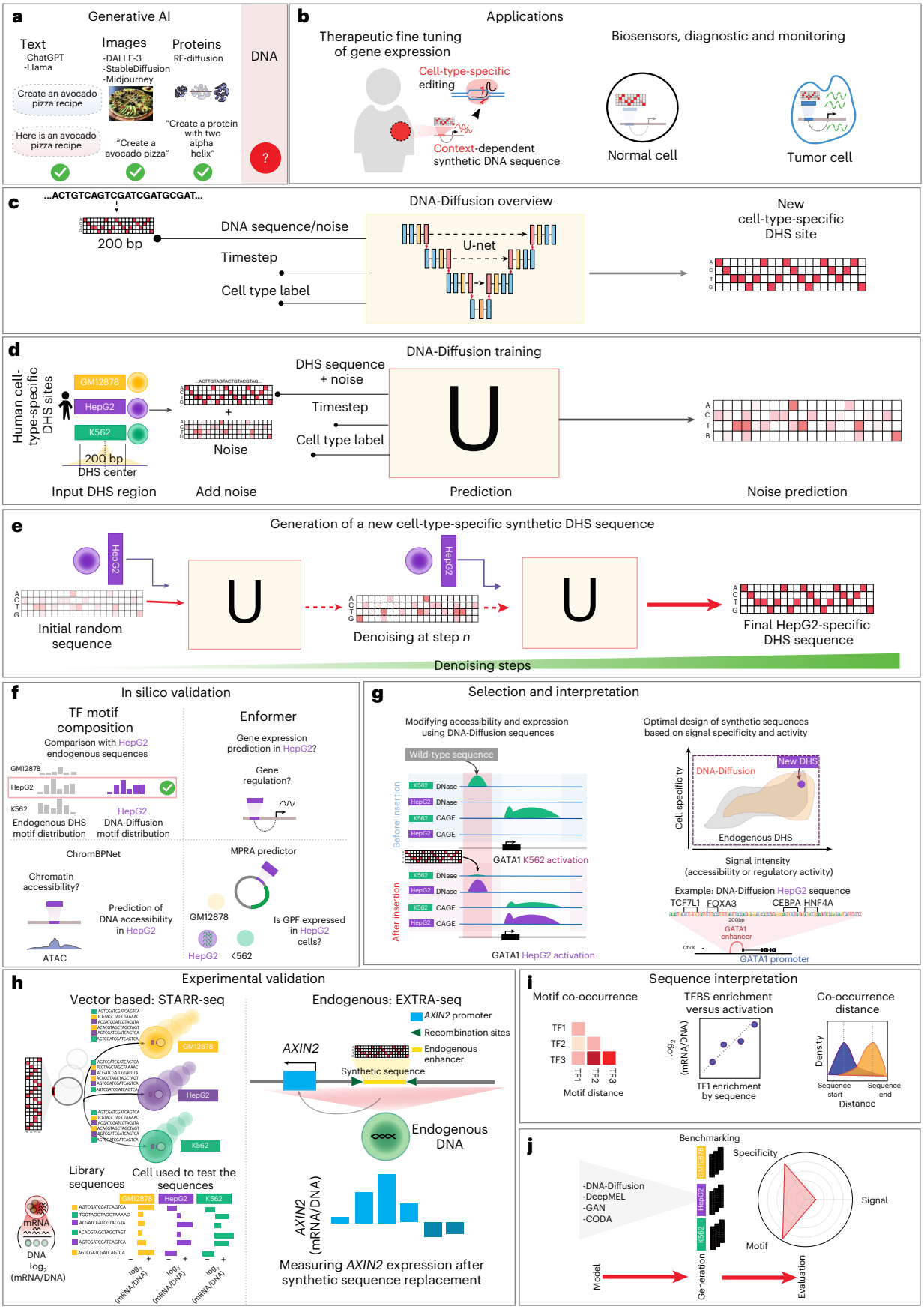

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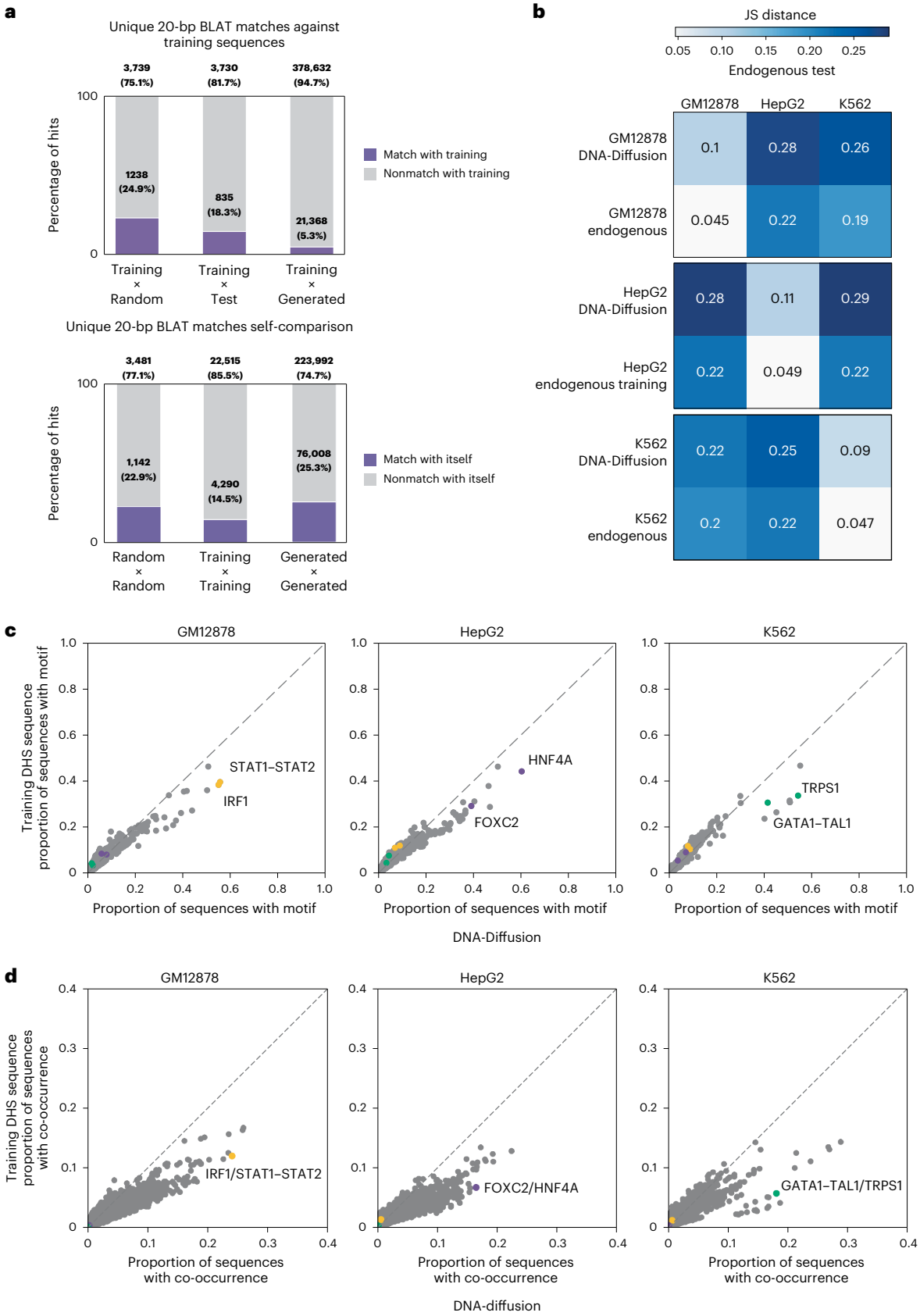

Having confirmed novelty, we assessed whether the sequences captured regulatory grammar using MOODS[34] to scan for TF binding motifs from the JASPAR[35] database. To compare the motif compositions, we calculated the Jensen–Shannon (JS) divergence between the motif distributions of generated sequences and the endogenous test set. This yielded a low mean JS divergence of 0.1 across the three cell types, revealing a high degree of similarity and suggesting that the model learned an appropriate regulatory vocabulary for each cellular

Fig. 2 | Sequence similarity and TF motif composition between DNA-Diffusion and endogenous regulatory elements. a, Bar plots showing the frequency (total number and percentage) of sequences with at least one alignment hit (minimum 20-bp match) against training sequences (top) or within the same group (bottom). In the top panel, sequences from different categories (randomly selected genomic coordinates, test endogenous DHS and DNA-Diffusion-generated) were aligned to the training set. In the bottom panel, each sequence group (randomly selected genomic coordinates, training endogenous DHS and DNA-Diffusion-generated) was aligned against itself. Purple bars indicate the percentage of sequences with at least one alignment hit, while gray bars represent the percentage without any detectable alignment against the target sequence set. **b**, JS distance comparing TF motif hit distributions between DNA-Diffusion sequences and endogenous train sequences against the endogenous test sets across GM12878, HepG2 and K562 cell types. **c**, A scatterplot showing the proportion of sequences with a given TF motif in endogenous training sequences versus DNA-Diffusion sequences across GM12878, HepG2 and K562 cell types. Important cell-type-specific TFs are annotated on the basis of existing literature (yellow, GM12878; purple, HepG2; green, K562). **d**, A scatterplot showing the proportion of sequences with a given TF motif pair co-occurrence in endogenous training sequences versus DNA-Diffusion sequences across GM12878, HepG2 and K562 cell types. Important cell-type-specific TFs are annotated on the basis of existing literature (yellow, GM12878; purple, HepG2; green, K562).

context (Fig. 2b). In fact, the model appeared to distill and enrich for the most informative cell-type-specific motifs. For example, HNF4A[36], a liver-specific pioneer factor, was present in approximately 47% of generated HepG2 sequences, compared with only 29% of endogenous HepG2 training sequences (Fig. 2c). We observed a similar enrichment for key immune-related factors such as IRF1 in GM12878 sequences (55% generated versus 38% training) and hematopoietic factors such as GATA1-TAL1[37] in K562 sequences (41% generated versus 31% training). Conversely, the model learned to exclude TF motifs specific to other cell types, suggesting it creates sequences with enhanced in silico specificity. This enrichment extended to TF motif co-occurrences, with IRF1-STAT1::STAT2 appearing in 24% of generated GM12878 sequences versus 12% in training, while preserving endogenous spacing and orientation (Fig. 2d, Supplementary Fig. 4 and Supplementary Note 6).

We discovered that the motif enrichment can be controlled by the classifier-free guidance (CFG) parameter[38], which enables tunable amplification of the conditioning signal during generation (Methods). By adjusting the CFG scale, we can modulate the density of these cell-type-specific TF motifs, consistent with the theory that diffusion models tend to concentrate probability mass in high-likelihood regions of the learned distribution[39]. Sequences generated without guidance (CFG scale 0) closely matched the training set's motif abundance, while a higher scale (CFG scale 2) produced even greater enrichment than the default (Supplementary Fig. 5). This finding demonstrates that diffusion naturally emphasizes functional determinants of cell-type specificity while de-emphasizing nonfunctional or shared elements, providing a powerful, interpretable lever to balance biological plausibility with functional optimization.

DNA-Diffusion-generated sequences demonstrate cell-type specificity, accessibility and cell-type-specific activation of *cis* genes in silico

To predict the functional potential of our synthetic sequences before costly experimental testing, we used state-of-the-art deep learning 'oracles' that predict genomic activity from sequence alone (Fig. 3a). We systematically replaced endogenous sequences at specific DHS sites with our generated sequences to determine if they could maintain or alter chromatin accessibility and gene expression in their respective cell types in silico.

We focused on regulatory regions of three well-characterized cell-type-specific genes: *GATA1* (K562), *HNF4A* (HepG2) and *CD19* (GM12878) (Supplementary Fig. 6 and Supplementary Table 1). Using ChromBPNet[40], which predicts chromatin accessibility at nucleotide resolution, we first tested sequences at a *GATA1* regulatory element previously validated by CRISPRi[41]. When K562-specific DNA-Diffusion sequences were inserted, they were predicted to maintain or enhance in silico accessibility specifically in K562 cells, achieving a significantly higher median *z*-scored normalized DNase I value (1.12) than training K562 sequences (0.92) ($P < 0.01$, Wilcoxon rank-sum test), while showing reduced activity in nontarget cells (GM12878: −0.62; HepG2: −0.74) (Fig. 3b). Conversely, GM12878- or HepG2-specific sequences induced in silico accessibility at this otherwise silent locus specifically in their target cell types, again producing a stronger signal than their training counterparts. These patterns held true when testing insertion at enhancers from other loci, such as those for *HNF4A* and *CD19* (Supplementary Figs. 7 and 8).

Next, to assess in silico transcriptional potential, we used a model trained to predict massively parallel reporter assays (MPRA) activity[42]. DNA-Diffusion sequences consistently showed high predicted cell-type-specific activity, with distributions that closely mirrored those of the endogenous training sequences (Fig. 3c), indicating the model learned to generate sequences with in silico transcriptional potential. We extended this analysis using Enformer[43], a powerful transformer-based model that predicts thousands of genomic tracks, including cap analysis of gene expression (CAGE), from a large 200-kb sequence context. Enformer's predictions for DNase I accessibility at the *GATA1* enhancer mirrored the ChromBPNet results (Fig. 3d). We then examined the predicted impact on in silico gene expression by looking at the Enformer CAGE signal near the *GATA1* transcription start site (TSS). The insertion of cell-type-specific DNA-Diffusion sequences at the validate enhancer site led to a significant increase in predicted promoter activity for each respective cell type ($P < 0.01$, Wilcoxon

Fig. 3 | Predictive modeling of in silico enhancer replacement using cell-type-specific DNA-Diffusion sequences. a, Left: strategy for replacing cell-type-specific sequences within putative enhancer regions of target genes (*GATA1*, *CD19* and *HNF4A*), along with predictive modeling of expected chromatin accessibility (ChromBPNet DNase I and Enformer DNase I) and gene expression regulatory potential (Enformer CAGE). Right: an approach using an MPRA predictor to evaluate the cell-type-specific expression potential of the sequences. **b**, Application of ChromBPNet to predict DNase I accessibility of cell-type-specific DNA-Diffusion-generated and endogenous sequences in the *GATA1* locus across GM12878, HepG2 and K562 cell lines. **c**, Predicted MPRA activity reflecting cell-type specificity in DNA-Diffusion and endogenous train sequences for GM12878, HepG2 and K562. **d**, Enformer DNase I predictions of chromatin accessibility at the *GATA1* locus, comparing cell-type-specific DNA-Diffusion and endogenous train sequences across GM12878, HepG2 and K562. **e**, Expression of the *GATA1* gene driven by both cell-type-specific DNA-Diffusion-generated and endogenous train sequences in GM12878, HepG2 and K562. For **b**–**e**, DNA-Diffusion sequences ($n = 100{,}000$) were used for all cell types. For DHS training sequences, n was 9,903 for GM12878, 9,833 for HepG2 and 9,769 for K562. All box plots display the median as the center line; the bounds of each box represent the interquartile range (IQR, 25th–75th percentiles); and whiskers extend to 1.5× the IQR beyond the box boundaries. Outliers beyond this range are shown individually where present. **f**, Top: DNase I and CAGE activity predictions at the *GATA1* locus based on the wild-type (WT) sequence. Bottom: post-insertion of a HepG2-optimized DNA-Diffusion sequence into the enhancer region, resulting in heightened chromatin accessibility and *GATA1* expression in HepG2, coupled with reduced accessibility in K562; GM12878 shows no substantial alterations. Chromatin accessibility profiles (DNase I) are represented in red, while gene expression measurements (CAGE) are depicted in blue throughout all panels. **g**, Comparative analysis of TF motifs in the WT and HepG2-specific DNA-Diffusion sequences of the *GATA1* regulatory region, with annotations of established cell-type-specific motifs from existing literature.

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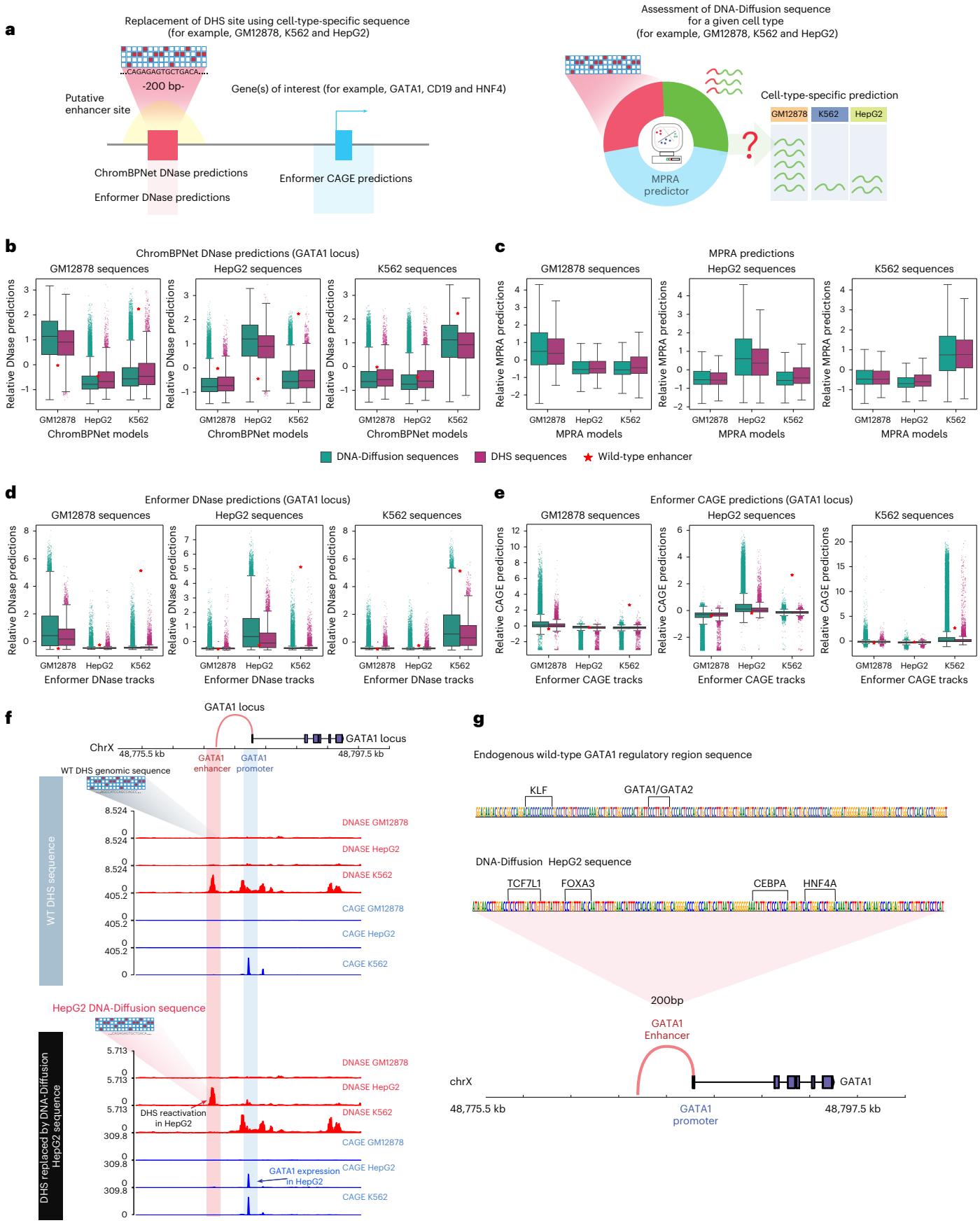

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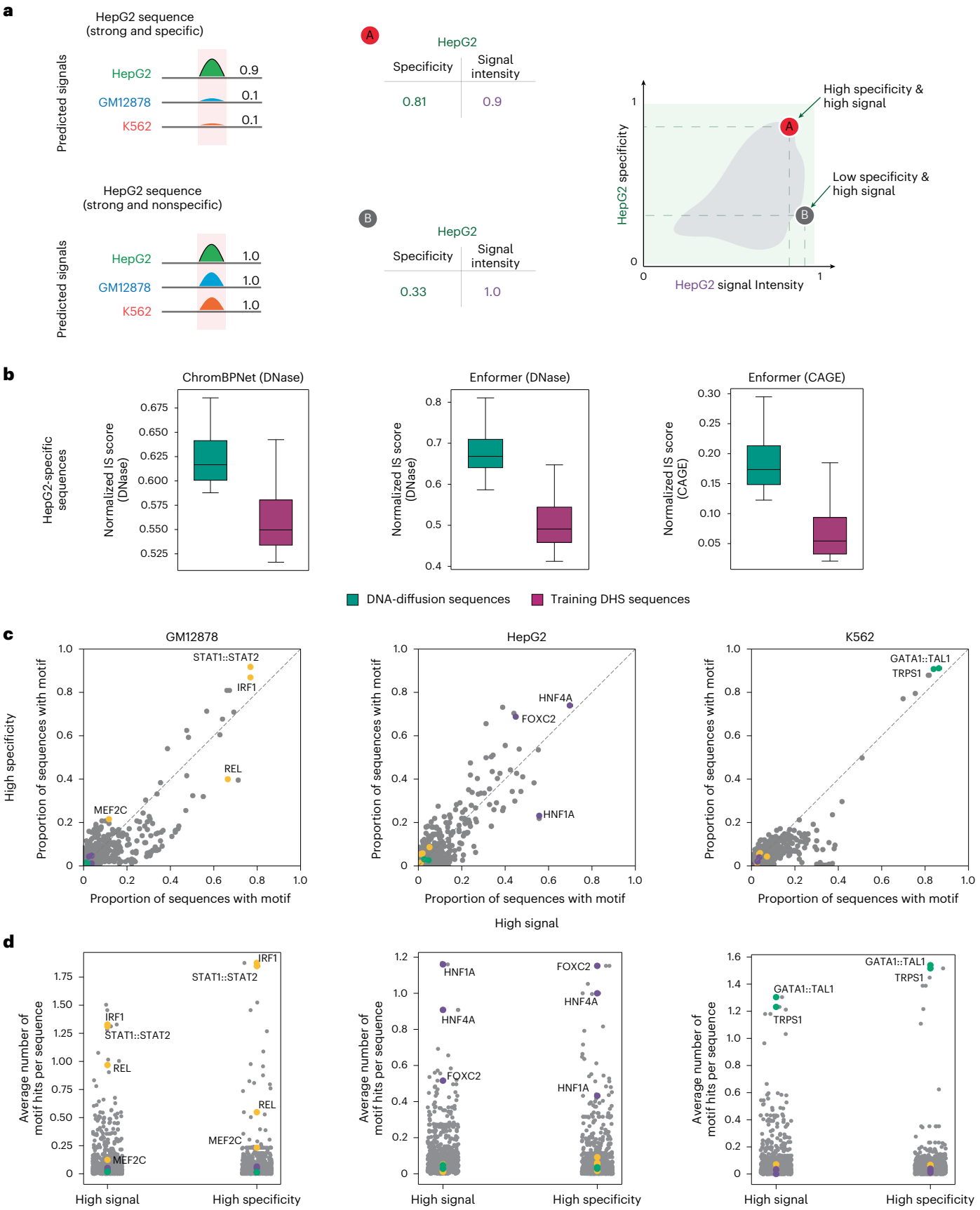

Fig. 4 | Evaluation of signal intensity versus signal specificity and their impact on downstream sequence composition. a, A schematic illustrating the workflow for evaluating both signal intensity and target cell specificity of regulatory sequences. Left: two hypothetical HepG2 sequences with their predicted signal profiles—one with high cell-type specificity (top, strong signal only in HepG2) and one with low specificity (bottom, strong signal in all cell types). Right: demonstratation of how these metrics can be visualized in a two-dimensional space, where the optimal sequences (high specificity and high signal) appear in the upper right quadrant. **b**, Box plots comparing the IS scores of the top 100 DNA-Diffusion sequences (teal) versus endogenous training sequences (purple) for HepG2-specific elements across three in silico oracles: ChromBPNet DNase I, Enformer DNase I and Enformer CAGE. DNA-Diffusion sequences consistently demonstrate higher median and maximum IS scores across all prediction models. All box plots display the median as the center line; the bounds of each box represent the interquartile range (IQR, 25th–75th percentiles); and whiskers extend to 1.5× the IQR beyond the box boundaries. **c**, Scatterplots showing the proportion of sequences containing specific TF motifs in high-signal (*x* axis) versus high-specificity (*y* axis) DNA-Diffusion sequences across three cell types. Key cell-type-specific TFs are highlighted on the basis of existing literature (yellow, GM12878; purple, HepG2; green, K562). **d**, Strip plots comparing the average number of TF motif hits per sequence between high-signal and high-specificity sequence sets across the three cell types. Cell-type-specific TF motifs are color-coded as in **c**.

rank-sum test), even in GM12878 and HepG2 cells, where the gene is normally inactive (Fig. 3e). Several generated sequences exceeded the in silico performance of both training data and wild-type enhancers (Fig. 3f) and contained known TF binding sites (Fig. 3g). Furthermore, using an in silico tiling strategy, we analyzed replacement at 200-bp windows within the *GATA1* locus and found an intronic region that significantly enhanced reactivation (Supplementary Figs. 9 and 10 and Supplementary Note 7).

Although we emphasize that creating an accessible region in silico does not necessarily indicate active gene expression, this analysis reveals a significant limitation of using Enformer as a predictor of gene expression. Using the *GATA1* enhancer as an example, in silico perturbation of its CRISPR interference (CRISPRi)-validated enhancer fails to reduce *GATA1* expression based on the K562 CAGE signal prediction (Fig. 3f). This limitation complicates efforts to examine potential off-target effects resulting from the introduction of new regulatory elements, consistent with previous studies documenting similar constraints[44,45].

Despite this limitation, these sequences consistently demonstrated predicted accessibility levels on par with or higher than DHS regions and were also able to induce the intended in silico cell-type-specific expression of the target genes. For loci with existing regulatory activity in specific cell types (for example, the distal *GATA1* enhancer in K562), we incorporated the wild-type enhancer sequence (marked with a star in Fig. 3b,d,e) as a reference point. Our analysis revealed that many synthetic sequences not only match but also exceed the chromatin accessibility and gene activation potential of the wild-type element across all computational oracles. Specifically, our DNA-Diffusion sequence set contains 483 GM12878 sequences, 507 HepG2 sequences and 553 K562 sequences that surpass the wild-type sequence across all in silico assays. This finding is particularly significant when contrasted with the training dataset, which includes only two GM12878, one HepG2 and one K562 sequence that outperform the wild-type sequence in K562. These synthetic sequences can activate *GATA1* to higher in silico levels than the endogenous wild-type sequence achieves in K562 and can do so even in cell types where *GATA1* is normally silent (HepG2 and GM12878). This pattern holds consistently across all three actively expressing loci (*GATA1*, *HNF4A* and *CD19*). This evidence suggested that our model could generate highly active and specific regulatory elements capable of precise gene modulation in silico.

Selection and characterization of synthetic sequences with different regulatory potential and cell-type specificity

The ability to generate a vast number of sequences enables the selection of elements with specific desired characteristics. We developed a framework to select DNA-Diffusion sequences along two dimensions: 'signal intensity' and 'signal specificity' (Fig. 4a). The product of these metrics yields a single intensity–specificity (IS) score (Methods). Given the complexity of using oracles to predict CAGE expression scores for genes that are already expressed in our tested cell lines, as demonstrated in our previous analysis of *GATA1*, *HNF4A* and *CD19*, we also analyzed *TNFSF18*. This gene presents both experimentally verified and computationally predicted low expression and low accessibility values across all three cell types while still providing differential prediction across all the target assays (DNase I and CAGE) (Supplementary Figs. 11 and 12). To validate these metrics, we compared our in-silico-derived scores with scores derived from experimentally measured chromatin accessibility data, finding strong correlations for signal intensity ($r = 0.57–0.67$), signal specificity ($r = 0.70–0.77$) and the combined IS score ($r = 0.65–0.74$ across multiple gene loci) (Supplementary Fig. 13 and Supplementary Table 3; Methods).

Using the IS score, we selected the top 100 sequences per cell type from both the DNA-Diffusion and training datasets at the unbiased *TNFSF18* locus (Supplementary Table 4). These results showed a compelling advantage of the synthetic sequences, which achieved higher IS scores across all three in silico oracles (ChromBPNet DNase, Enformer DNase and Enformer CAGE) compared with the best training

Fig. 5 | STARR-seq validate cell-type-specific activity of DNA-Diffusion sequences. a, A schematic of the STARR-seq experimental design illustrating the comprehensive evaluation workflow. Left: library construction combining DNA-Diffusion synthetic sequences and endogenous regulatory elements from three cell types (K562, HepG2 and GM12878). Middle: the pooled library is transfected into all three cell lines, with each plasmid carrying a unique sequence. Right: regulatory activity is quantified by measuring the ratio of self-transcribed mRNA to plasmid DNA abundance for each sequence. **b**, Box plots display the distribution of normalized $\log_2$(RNA/DNA) ratios for eight sequence categories in K562 (left), HepG2 (middle) and GM12878 (right). Groups include: (i) DNA-Diffusion sequences with both high predicted signal and specificity ($n = 100$ per cell type); (ii) sequences with high signal only; (iii) sequences with high specificity only; (iv) all DNA-Diffusion sequences tested ($n = 1{,}234$ GM12878, 1,010 HepG2 and 992 K562); (v) model-sampled sequences without predictor bias; (vi) endogenous DHS; (vii) experimentally validated positive controls; and (viii) negative controls with low predicted activity. The *x* axis represents the *z*-scored STARR-seq $\log_2$(RNA/DNA) ratio relative to endogenous training sequences, while the *y* axis lists the sequence categories. Colors correspond to sequence origin (K562, green; HepG2, purple; GM12878, yellow). Each box shows the median, interquartile range (IQR) and whiskers extending 1.5× the IQR. **c**, Correlation between normalized Enformer DNase I IS scores (*x* axis) and experimental STARR-seq $\log_2$(RNA/DNA) ratios (*y* axis) across K562, HepG2 and GM12878 cell lines. Each point represents an individual sequence, with correlation coefficients shown for each cell type. ρ represents the correlation coefficient between DNase I and STARR-seq. **d**, GSEA showing examples of different TFs for the STARR-seq performed in different cell types. The panels show TFs positively enriched in sequences with high RNA/DNA ratios. The *x* axis shows the sequence rank of the endogenous *z*-scored $\log_2$ (RNA/DNA) of all the DNA-Diffusion sequences present in the STARR-seq library in the three tested cell lines. The black lines represent the TF presence in a sequence. The green curve on the *y* axis shows the normalized enrichment score (NES) for each TF. Statistical significance was assessed with the GSEA method (Methods). FDR, false discovery rate.

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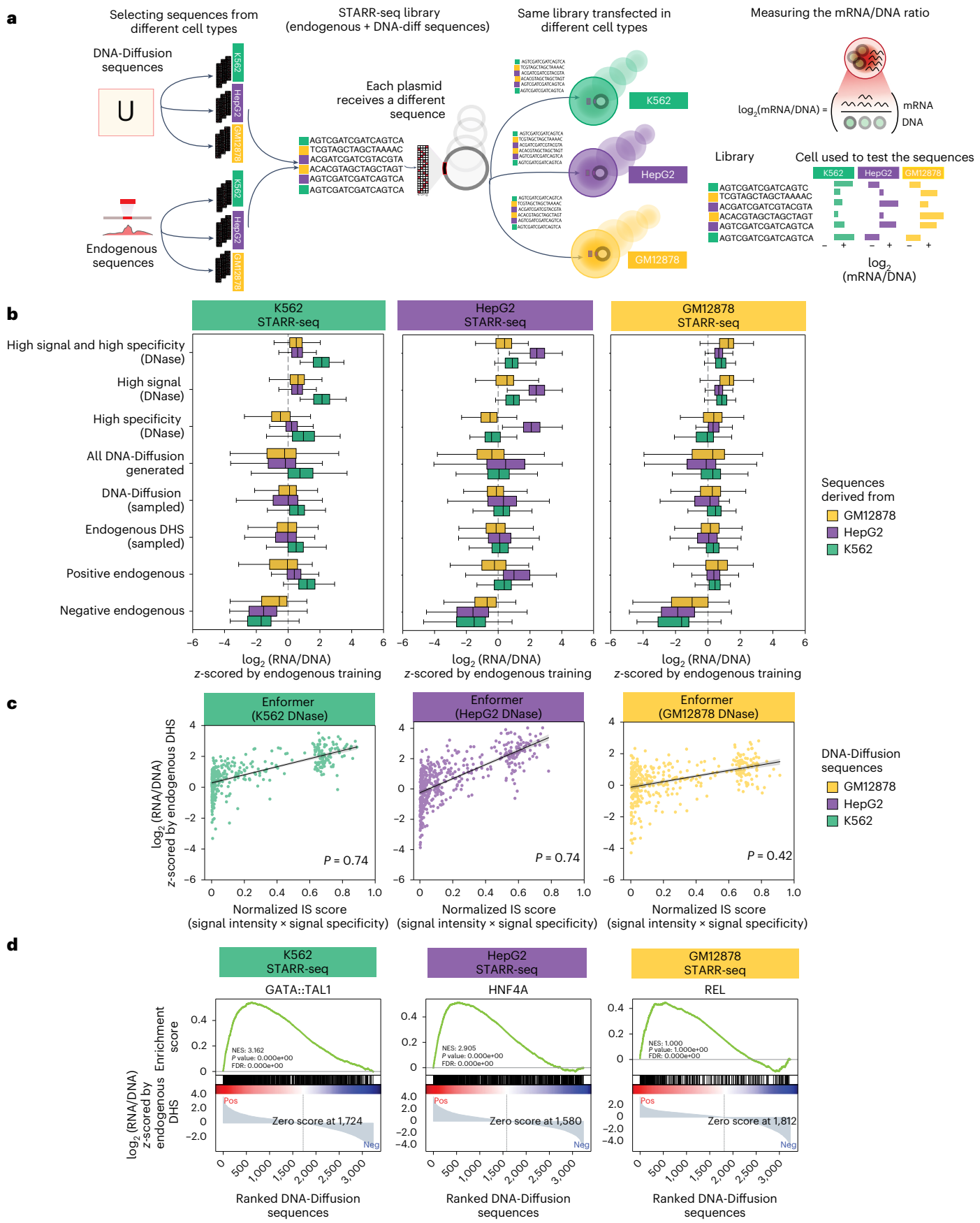


DHS sequences (Fig. 4b and Supplementary Fig. 14). This highlights the model's capability to discern sequence features unique to specific cell types and generate sequences that optimize both signal intensity and specificity beyond what is typically found in the genome.

We then investigated the TF motif composition of sequences classified as high signal versus high specificity (Fig. 4a). In HepG2 sequences, the HNF1A motif was enriched in high-signal sequences (56% versus 23% in high specificity), while the FOXC2 motif was enriched in high-specificity sequences (69% versus 45% in high signal), suggesting a learned regulatory logic for tuning these properties (Fig. 4c,d). Similar patterns were observed in GM12878 with immune-related TF motifs such as REL and MEF2C and GATA family motifs in K562. TF expression analysis revealed that many motifs enriched in high-specificity and high-signal sequences are predominantly expressed in a cell-type-specific manner (Supplementary Fig. 15). These findings suggest a sequence-based regulatory logic distinguishing these two categories of sequences, underscoring the nuanced relationship between TF motifs and their regulatory impact in specific cellular contexts.

Using STARR-seq to evaluate the regulatory activity of DNA-Diffusion sequences

To validate the predictions obtained for the DNA-Diffusion sequences, we performed STARR-seq[27,28] on 5,850 DNA elements in K562, HepG2 and GM12878 cells to measure the transcriptional activity of each element in an episomal context (Fig. 5a).

STARR-seq quantitatively measures the ability of DNA sequences cloned into reporter plasmids to drive transcription, determined by the ratio of the abundance of mRNA transcripts derived from the DNA fragments relative to their representation in the original DNA pool (Fig. 5a). Although this assay does not directly probe chromatin accessibility, its quantitative output has been shown to correlate positively with accessibility data[46], validating its suitability for evaluating the functional activity of DNA-Diffusion sequences. Our library design included multiple endogenous control sets and DNA-Diffusion sequences selected using various strategies (Supplementary Table 5e and Supplementary Note 8).

The results robustly confirmed our computational predictions despite the known shortcomings of the predicted oracles we used[44,45]. Endogenous DHS sequences established experimental baselines, showing higher activity in their native cellular contexts. Randomly sampled DNA-Diffusion sequences accurately mirrored this behavior, demonstrating that the model captures the quantitative regulatory activity of the sequences it was trained on (Fig. 5b). DNA-Diffusion sequences selected for high in silico intensity or a high IS score outperformed strong, previously validated positive control elements. Despite the DNA-Diffusion model not being trained on expression data, the top-ranked generated sequences consistently outperformed the strongest validated genomic sequences experimentally. Importantly, incorporating specificity in addition to signal intensity for the selection of generated sequences did not compromise their transcriptional potential. STARR-seq activity correlated strongly with in silico Enformer DNase I scores, with Pearson correlation coefficients reaching 0.74 in both K562 and HepG2 (Fig. 5c). The lower correlation in GM12878 (0.42) was consistent with known transfection efficiency challenges in this cell line[47].

To link function to sequence grammar, we performed gene set enrichment analysis[48] (GSEA) on TF motifs within a sequence set ranked by their experimentally observed regulatory activity levels (Supplementary Table 6; Methods). This confirmed that high transcriptional activity was significantly associated with the presence of the expected cell-type-specific TF motifs (for example, GATA1-TAL1 in K562, HNF4A in HepG2 and REL in GM12878) (Fig. 5d and Supplementary Figs. 16 and 17).

These results underscore DNA-Diffusion's potential to generate functionally active, cell-type-specific regulatory elements beyond naturally occurring genomic counterparts. Our generative framework enables efficient selection of sequences demonstrating high regulatory activity and specificity, significantly expanding the repertoire of experimentally validated regulatory elements for diverse biological applications.

DNA-Diffusion sequences can reactivate a human leukemia protective gene in an endogenous context

Building upon our previous findings that highlighted DNA-Diffusion's capability to generate cell-type-specific chromatin accessibility and precisely modulate gene expression both computationally and experimentally, we sought to test its efficacy within an endogenous cellular context. Specifically, we targeted the locus of Axin-related Protein 2 (*AXIN2*), a gene often silenced in leukemic B cells. Interestingly, a naturally occurring protective genetic variant (rs143348853), located approximately 2.5 kb upstream of the *AXIN2* TSS within a distal regulatory element, can reactivate *AXIN2*. This variant enhances the binding affinity of MEF2-family TFs, modestly increases *AXIN2* expression and, importantly, correlates with significantly improved prognosis in chronic lymphocytic leukemia[30] (Fig. 6a,b).

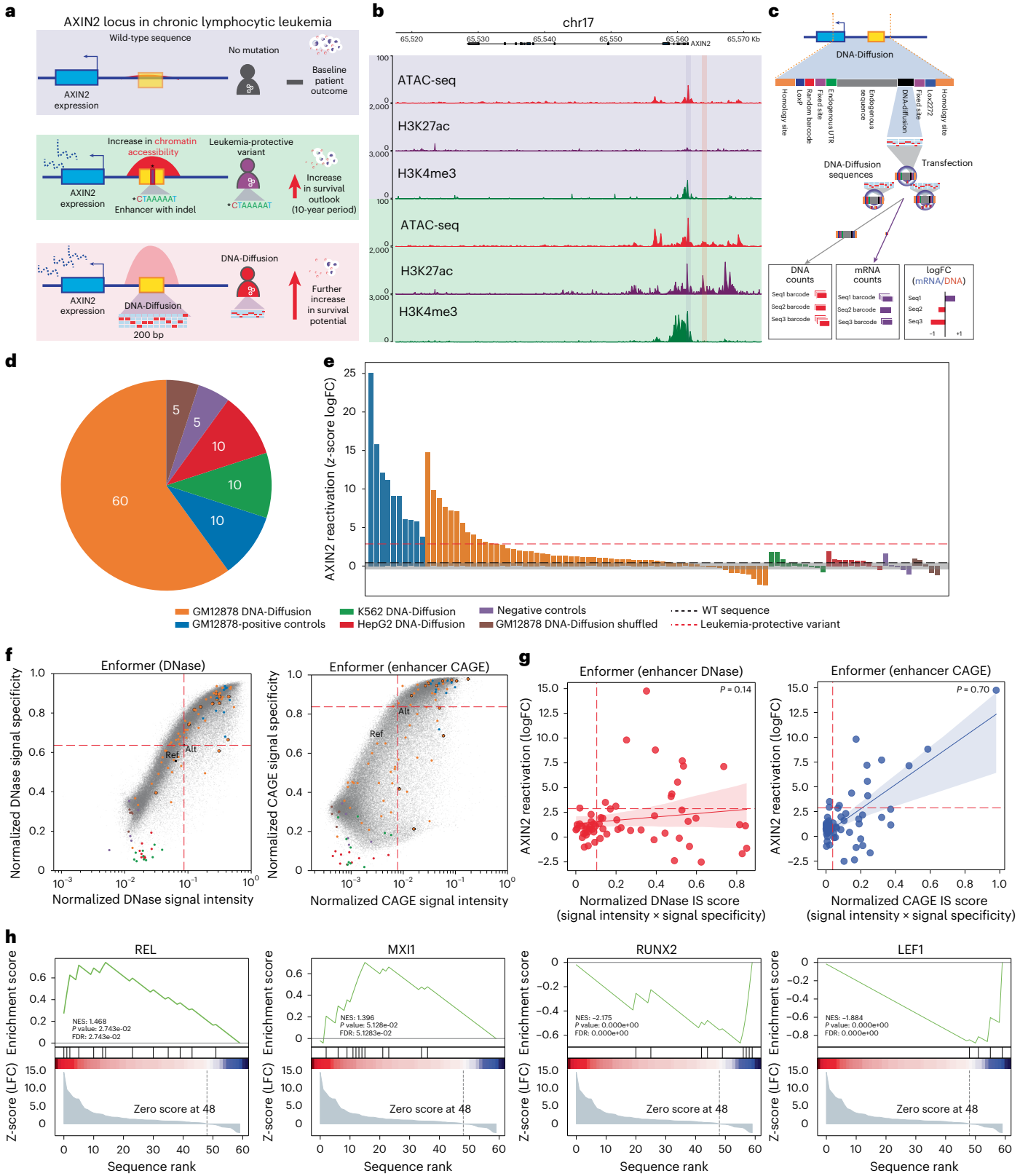
To evaluate the capacity of DNA-Diffusion sequences to endogenously modulate *AXIN2* transcription, we used a MPRA-like system, called EXTRA-seq[3], which utilizes CRE-recombinase mediated cassette exchange and long-read sequencing to measure the gene's transcriptional output in response to enhancer sequences (Fig. 6c and Supplementary Table 7a–d).

Fig. 6 | DNA-Diffusion sequences enhance *AXIN2* expression beyond protective variant levels. a, A schematic depicting the wild-type sequence lacking upstream regulatory regions and low *AXIN2* expression (top), a 5-bp indel creating a putative regulatory region that enhances *AXIN2* expression and increases survival (middle), and the introduction of DNA-Diffusion to boost patient survival outcomes by inducing *AXIN2* expression greater than the protective variant (bottom). **b**, ATAC-seq, H3K27ac and H3K4me3 tracks for two cell lines derived from two different individuals without the indel (top tracks) and one with the indel (bottom tracks). **c**, A schematic outlining the experiment introducing DNA-Diffusion sequences endogenously to the MEC-1 cell line to examine their effect on *AXIN2* expression using a CRE-recombinase-based system. **d**, Composition of sequences used in the experimental validation. **e**, Measured $\log_2$FC (fold change) of *AXIN2* reactivation across validation sequences, with sequences colored according to the legend. Log_2FC of *AXIN2* reactivation was calculated as the log ratio between the mRNA read counts (from different sequences barcode-tagged *AXIN2* transcript) and the genomic DNA (gDNA) read counts for the same barcodes. The red dashed line represents the $\log_2$FC of the leukemic protective variant. **f**, Scatterplots of Enformer DNase I and CAGE signal intensity versus specificity for GM12878 DNA-Diffusion sequences. Validation sequences are colored per the legend; sequences exceeding the alternative variant's $\log_2$FC are outlined in black. The dashed red line denotes Enformer predictions for alternative sequence. **g**, Correlation of the Enformer DNase I and enhancer CAGE predictions to *AXIN2* reactivation measured by the $\log_2$FC *z*-scored (standard score) relative to the negative sequences' mean and standard deviation. The dashed red line depicts the normalized Enformer predictions and measured $\log_2$FC for the alternative sequence. Shaded areas around regression lines indicate 95% confidence intervals for the fit regression estimate, as computed by seaborn.regplot with default parameters. ρ represents the correlation coefficient between *AXIN2* reactivation (EXTRA-seq) and Enformer predictions. **h**, A selection of GSEA for four TF motifs where the two on the left are enriched in sequences that present higher $\log_2$FC than the alternative sequence, while the two on the right are enriched in sequences that present lower $\log_2$FC than the alternative sequence. **i**, Left: DNA-Diffusion sequence that presented the highest $\log_2$FC with GM12878-specific TFs and GSEA TFs annotated. Right: DNA-Diffusion sequence that presented the lowest $\log_2$FC with GSEA TFs annotated. Statistical significance was assessed with the GSEA method (Methods).

https://doi.org/10.1038/s41588-025-02441-6

We tested a library of 100 sequences in a B cell leukemia cell line (MEC-1), including 60 GM12878-specific DNA-Diffusion sequences selected to span a range of predicted activities, along with various positive and negative controls (Fig. 6d; Methods). DNA-Diffusion sequences designed for GM12878 effectively reactivated endogenous *AXIN2* expression, with multiple sequences surpassing the activity of the natural protective variant and achieving activation levels comparable to strong positive controls from other studies (Fig. 6e; Methods). Sequences designed for non-B-cell contexts (K562 and HepG2) showed no significant activity, confirming the cell-type specificity of the synthetic sequences in an endogenous setting.

For practical applications where it is necessary to select and validate a small set of sequences with a given design goal, we assessed which computational metrics best predicted functional outcomes.



https://doi.org/10.1038/s41588-025-02441-6

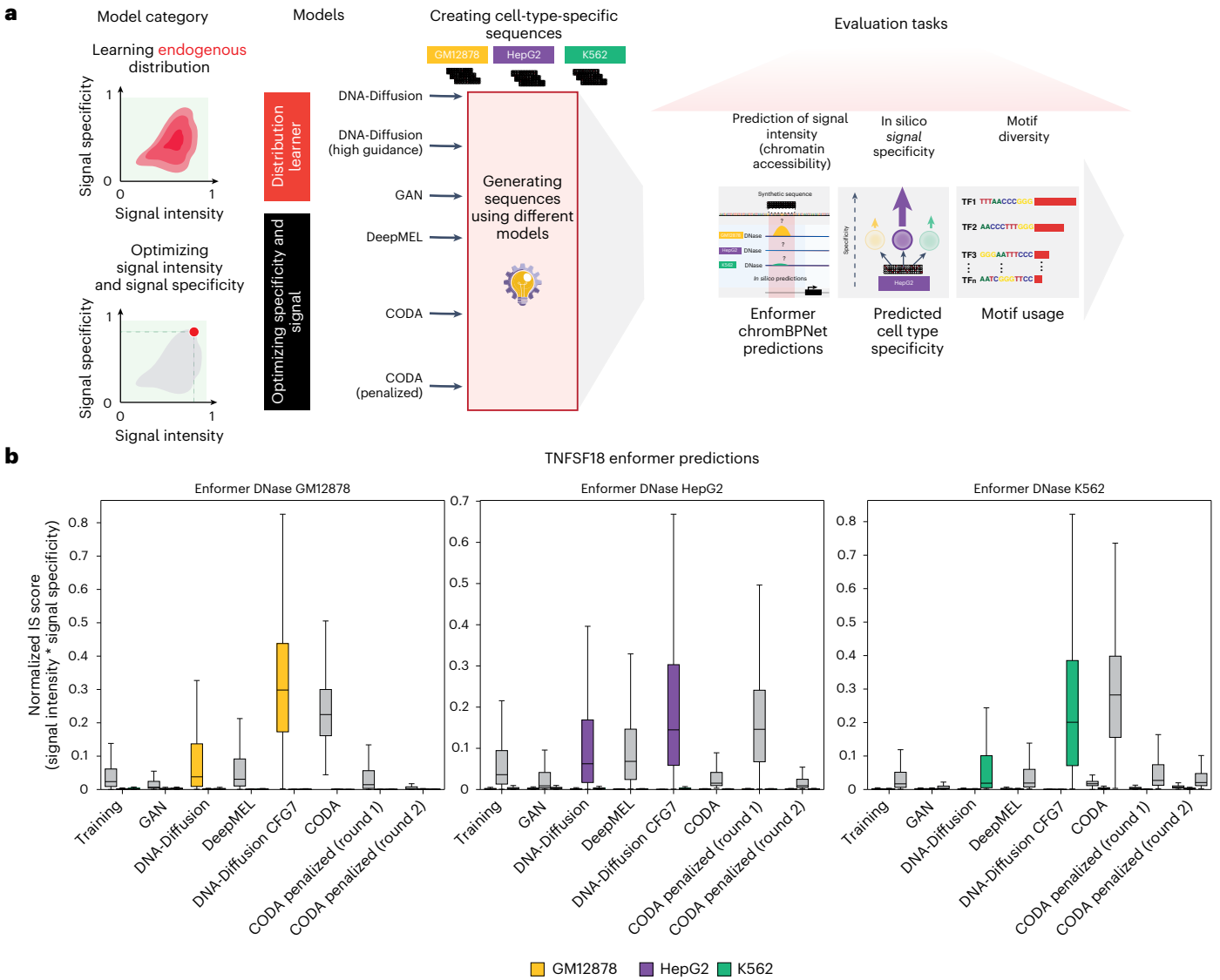


Fig. 7 | Benchmarking current state-of-the-art synthetic regulatory element deep learning tools. a, A schematic comparing different model categories: distribution learners (DNA-Diffusion and GANs) that learn from endogenous sequences versus optimizers (DeepMEL and CODA) that directly maximize signal intensity and specificity of a sequence based on a pretrained predictor. The evaluation framework assesses all models on three key metrics: signal intensity (chromatin accessibility), signal specificity (cell-type specificity) and motif diversity (TF motif composition). **b**, Box plots showing Enformer DNase I IS score performance across GM12878, HepG2 and K562 assays at the *TNFSF18* locus with cell-type-specific sequences highlighted in their corresponding colors (yellow, GM12878; purple, HepG2; green, K562) (n = 1,000 for each sequence category and cell type). All box plots display the median as the center line; the bounds of each box represent the interquartile range (IQR, 25th–75th percentiles); and whiskers extend to 1.5× the IQR beyond the box boundaries. **c**, A comparison of motif diversity scores (1 minus Gini coefficient) across all methods, where higher values indicate more diverse TF motif representation. Note the trade-off between performance in **b** and diversity in **c**, particularly for high-guidance models. **d**, Radar plots providing unified visualization of each model's performance across all three metrics, with the normalized area score (model score/maximal possible score) shown in the center. DNA-Diffusion plots (highlighted in red) demonstrate superior balance across all dimensions compared with other methods (gray).

CAGE signal intensity was the best-ranking metric for selecting sequences, with six out of the top ten sequences exhibiting $\log_2$fold change (FC) greater than the protective variant and yielding results comparable to the positive controls. The CAGE IS score (enhancer site) performed similarly, identifying five of its top ten ranked sequences that met this threshold. Remarkably, all top three sequences ranked by CAGE IS score group strongly reactivated *AXIN2* (Fig. 6f and Supplementary Fig. 18).

Post-hoc analysis revealed that the Enformer prediction of CAGE signal at the enhancer insertion site had the strongest correlation to measured transcriptional activity, with a Spearman rho of 0.70 (Fig. 6g and Supplementary Fig. 19). DNase I accessibility predictions showed little to no correlation, suggesting that chromatin accessibility alone is not a strong predictor of activation potential in this context.

Furthermore, we discovered motifs enriched in sequences that, despite showing high levels of predicted chromatin accessibility, failed to reactivate *AXIN2*. These factors include Runt-related transcription factor 2 (RUNX2), known for its role in osteoblast differentiation[49] and lymphoid enhancer binding factor-1 (LEF1), which mediates the Wnt pathway in chronic lymphocytic leukemia (CLL) cells[50]. RUNX2 upregulation has been shown to act as a repressor in patients with acute lymphocytic leukemia and plays a key role in disease progression[51] (Fig. 6h). This finding suggests that factors beyond chromatin accessibility, such as the presence of other regulatory elements or the recruitment of additional transcriptional machineries, may be necessary to drive *AXIN2* expression.

These findings demonstrate the potential of DNA-Diffusion to design sequences with therapeutic potential, showing their effectiveness in an endogenous setting. By identifying TFs that are more likely to be effective and optimizing the generation to incorporate or exclude specific TF 'words', we can fine-tune DNA-Diffusion sequences for enhanced performance. This optimization approach opens exciting opportunities for targeted gene regulation and the development of therapeutic strategies.

Benchmarking DNA-Diffusion against state-of-the-art methods for the generation of synthetic regulatory elements

We systematically benchmarked DNA-Diffusion against other leading approaches for regulatory element design including generative adversarial networks (GANs), which synthesize sequences by iteratively improving a generator through adversarial feedback from a discriminator and convolutional neural networks (CNN) based regressors[17,22,52]. We retrained these models on our dataset and generated 1,000 sequences per cell type for each model (Methods; Supplementary Note 9). We developed an in silico validation framework to evaluate all methods on three key metrics: signal intensity, cell-type specificity and motif diversity (Fig. 7a; Methods). For signal intensity and specificity assessment, we used Enformer DNase I predictions at the *TNFSF18* locus, chosen to minimize potential bias from pre-existing cell-type-specific expression patterns. To quantify motif diversity, we developed a metric based on the Gini coefficient, where a score of 1 indicates highly diverse motif usage.

Results revealed fundamental trade-offs between generation approaches. Optimization-focused approaches, such as CODA, generate sequences with higher predicted activity (average median IS score of 0.168), but this comes at the cost of reduced motif diversity (average median score of 0.24), as the model converges on a few optimal TF motifs (Fig. 7b,c). CODA also incorporates a motif penalization procedure that enables enhanced sequence diversity generation, with sequential rounds yielding progressively degrading IS scores: round 1 (0.065) and round 2 (0.001). However, this penalization requires manual, cell-type-specific tuning and expertise in TF motif composition.

By contrast, standard DNA-Diffusion (with moderate CFG of 1) produced a more balanced output (IS score 0.04, diversity 0.53), outperforming GANs (IS score 0.006, diversity 0.59) and matching the activity of DeepMEL (IS score 0.041, diversity 0.58) while maintaining comparable diversity to endogenous elements (IS score 0.03, diversity 0.58). Increasing CFG to 7 tuned DNA-Diffusion to outperform all other methods on signal intensity and specificity (IS score 0.215), effectively operating as a powerful sequence optimizer when desired, albeit with a corresponding drop in diversity (diversity score 0.36). This allows us to bridge the gap between sequences that resemble endogenous elements and those designed to maximize activity.

To systematically evaluate how these metrics translate to practical sequence selection for experimental validation, we examined model performance under simulated validation conditions. We selected and analyzed each model's top-performing sequences (100 per cell type) based on minimal Euclidean distance from the optimal point (1, 1) in the predicted signal intensity/specificity space and their motif diversity (Supplementary Table 8 and Supplementary Note 10).

A unified performance metric, represented by the area of a radar plot summarizing all three axes, confirmed that standard DNA-Diffusion offers the best overall balance of activity, specificity and motif diversity (area score 0.71) while DNA-Diffusion with CFG7 had the highest area score overall (0.84) (Fig. 7d).

Taken together, DNA-Diffusion distinguishes itself through its versatility, enabling the generation of sequences that closely mirror endogenous sequence patterns or optimized for activation. While CODA and DeepMEL can generate sequences that exceed the in silico signal potential of their training data, DNA-Diffusion's tunability via CFG allows it to excel across different design objectives without requiring manual intervention for each cell type.

Discussion

We developed DNA-Diffusion, a generative AI framework that designs compact regulatory elements with tunable cell-type specificity. Through experimental validation, we demonstrated that these designed sequences can modulate *AXIN2*, a disease-relevant gene, in its native genomic context, suggesting potential therapeutic applications.

DNA-Diffusion differs from prior methods in several ways. Consistent with theoretical predictions[53], the model learns to generate potent elements in an end-to-end fashion, without requiring manual cell-type-specific tuning. CFG allows users to balance the generation of endogenous-like elements with sequences designed for maximum activation potential. Benchmarking revealed this versatility is notable,

as DNA-Diffusion achieves a superior balance of signal intensity, specificity and motif diversity, increasing the likelihood of generating functionally robust elements with minimized off-target potential.

High-throughput STARR-seq assays confirmed that DNA-Diffusion can generate several elements that are more active and specific than endogenous controls. More importantly, the EXTRA-seq experiment at the *AXIN2* locus provides a proof of concept for the therapeutic potential of this technology. By demonstrating that computationally designed sequences can reactivate a protective gene in a disease-relevant context, outperforming a natural genetic variant, this work provides a foundation for precision gene therapies.

This denoising diffusion probabilistic model (DDPM)-based DNA diffusion model has been validated with both large-scale reporter assays and direct modification of an endogenous locus. While this marks a substantial advance for the field, our model has limitations. Because DNA-Diffusion was trained to reproduce the accessibility distribution and diversity of endogenous DHS regions, our approach requires a second step to select sequences with higher regulatory activity using oracles. As shown in Fig. 5b, randomly sampled DNA-Diffusion sequences exhibit similar activities to endogenous DHS elements, with enhanced signals emerging only after computational preselection. In addition, its reliance on binarized labels for DHS peaks could be improved by incorporating continuous measures of chromatin activity. Recent diffusion models and large language models have explored this idea and have demonstrated promise in in silico studies, warranting experimental validation to confirm their efficacy[19,21,54–56]. Finally, we expect this approach will benefit from extension to a broader range of cell types and from creating active learning cycles where experimental results are used to progressively refine the model.

In conclusion, DNA-Diffusion is a robust platform that holds immense promise for developing gene therapies and deepening our understanding of the DNA regulatory code and grammar.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-025-02441-6.

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Methods

Data preprocessing for training

We selected training data from an annotated index of DHS regions in the human genome[32], focusing on data for K562, GM12878 and HepG2 cell lines based on ease of laboratory culture, tissue diversity and availability of additional annotation and data. A detailed description of the curation and filtration methods is provided in Supplementary Methods 1.

Model architecture

Our DNA-Diffusion model extends the foundational DDPM framework[8], with architectural implementations informed by the Annotated Diffusion Model. We implemented several fundamental adaptations to optimize the architecture for nucleotide sequence generation. In contrast to conventional image-based diffusion models that operate on square inputs and pixel values, our model processes DNA sequences using a dimensional tensor structure of (batch, 1, 4, 200), wherein the dimension of 4 corresponds to the four nucleotides (A, C, T and G) and 200 represents the sequence length. The U-Net architecture preserves the essential structural elements from the original DDPM framework while reconfiguring channel dimensions specifically for DNA sequence representation, projecting encoded sequences to dimensions that optimally balance nucleotide dependency modeling with computational efficiency. We provide a more detailed description of the components of the U-Net backbone in Supplementary Methods 2.

For model training, we implemented cell-type-specific conditioning with DNA-appropriate modifications to enable targeted regulatory sequence generation. The conditioning protocol incorporated cell-type labels with a 10% random masking rate during training to facilitate CFG during the sampling phase. These conditioning labels were concatenated with timestep embeddings and integrated into the ResNet[57] blocks, while attention layers[58,59] exclusively processed the noised input sequences. During the reverse diffusion process, conditional and unconditional samples were jointly processed with a guidance scale parameter of 1.0. Through this methodological approach, we generated a comprehensive dataset of 100,000 sequences per cell type, yielding a total of 300,000 generated regulatory sequences for subsequent analysis and validation. We note that utilizing a guidance scale of 1 generates class conditional outputs with moderate guidance and 0 generates class-conditional outputs with no additional guidance.

Training process

The main diffusion model was implemented in PyTorch 2.0.0 and trained on four 40 GB NVIDIA A100 graphics processing units with a batch size of 960 for 2,000 epochs, requiring approximately 10 h to complete. Subsequent sequence generation using a single PCIE A100 graphics processing unit proceeds at a rate of approximately 15 sequences per second. For distributed computing across the nodes, we used Slurm Workload Manager. We used the Adam[60] optimizer with a learning rate of 1×10^{-4} along with PyTorch default values for other hyperparameters: $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The linear noising schedule was used with $\beta_{start} = 0.0001$ and $\beta_{end} = 0.005$.

In silico validation

Sequence similarity search. We performed the BLAT[33] similarity search using two strategies. In the first strategy, we used the training sequences as the database and searched for matches using the generated, test and randomly selected genomic coordinate sequences. The random sequences were obtained using the BedTools[61] random function, which selects genomic regions without bias toward their sequence composition. After the initial selection, we removed sequences intersecting with DHS sites used in our training and test datasets.

For the analysis that included only alignments without overlap with genomic repetitive elements, we used all repetition types in the hg38 RepeatMasker[62,63] available in the UCSC Genome Browser database to filter out hits that intersected with the training DHS regions. This intersection was created using the command bedtools intersect with a bed file containing the alignments coordinates regions relative to the train sequences and another file with repetitive regions.

In the second strategy, we considered alignments within groups using the same sequence set as both the database and the query (for example, Train × Train, Random × Random, and Generated × Generated). The BLAT parameters used in both strategies were: q=dna, tileSize=11, stepSize=5, -minMatch=1, repMatch=2253, minIdentity=100, noHead=True and minScore=20 or 30.

Motif scanning. To scan motifs, the MOODS[34] 1.9.4.1 package was used using the vertebrate motifs from the JASPAR 2024 position weight matrices (PWMs) database[35] (n = 880 PWMs). The parameters used in MOODS were a P value of 0.0001 and a pseudocount (--ps 0.0001). We observe the appearance of key cell-type-specific TF motifs around step 30 that resolve more granularly until the final timestep 50 (Supplementary Fig. 20). We provide all scanning (Supplementary Data 1) and co-occurrence (Supplementary Data 2) results as supplementary data.

Assessing DNA chromatin accessibility with ChromBPNet. Chromatin accessibility is a hallmark of regulatory activity for a DNA sequence. We used the ChromBPNet[40] model as a biological oracle to assess the cell-type specificity of generated diffusion sequences' chromatin accessibility. Details of ChromBPNet's architecture are noted in Supplementary Methods 3.

We used ChromBPNet models trained on ENCODE[1,2,4] experiment data ENCSR000EPC, ENCSR000EMT and ENCSR000ENP to predict base-pair chromatin accessibility of DNA-Diffusion sequences in K562, GM12878 and HepG2 contexts, respectively. Each ChromBPNet model's input was a one-hot encoded 2,114-bp DNA sequence (A = [1,0,0,0], C = [0,1,0,0], G = [0,0,1,0], T = [0,0,0,1]).

To evaluate how DNA sequences generated by the diffusion model affect chromatin accessibility, we inserted them within known or putative regulatory regions for genes *GATA1*, *CD19*, *HNF4A* and *TNFSF18* and calculated the mean of the predicted base-resolution chromatin accessibility score.

For a generated sequence designed for a particular cell type, we expect that the ChromBPNet model trained to predict accessibility in that cell type will yield higher chromatin accessibility predictions than the other two ChromBPNet models, indicating greater regulatory potential. However, the three ChromBPNet models used were trained on experimental data with different read depths: ENCSR000EPC had a read depth of 245 million, ENCSR000EMT had a read depth of 68 million and ENCSR000EMT had a read depth of 323 million. Thus, we observed that models trained on deeper data typically predicted higher chromatin accessibility compared with the other two models for a generated sequence, irrespective of the cell-type specificity of a sequence—leading to inaccurate comparison of model outputs. To combat this potential confounder, we applied quantile normalization (see 'Normalization and scaling of the data' section).

MPRA predictor. We trained a model[42] to predict MPRA activity using MPRA data from GM12878, K562, HepG2, SK-N-SH and A549 cells (accession IDs are available in Supplementary Table 3). The assayed sequences are 200 bp long, and most of them are derived from human genomic segments containing the reference and alternate alleles for variants from UK Biobank[64] and GTEx[65]. These sequences are cloned upstream of a reporter gene, and this construct is delivered to host cells using transient transfection. Each sequence's MPRA activity is measured as the $\log_2$ of the ratio of the number of mRNA molecules produced using the construct to the number of constructs delivered to the host cells. From ENCODE, we obtained 318,734, 636,185, 750,298, 750,084 and 318,734 MPRA activity measurements from GM12878, K562, HepG2,

SK-N-SH and A549 cells, respectively. The training procedure of the MPRA predictor is described in Supplementary Methods 4.

The trained models were accurate and achieved Spearman's rank correlation coefficients of 0.4527, 0.7285, 0.7682, 0.7745 and 0.6682 on the test sets for GM12878, K562, HepG2, SK-N-SH and A549 cells, respectively. They also achieved Pearson's correlation coefficients of 0.4479, 0.8115, 0.8257, 0.8199 and 0.7218 on the test sets for GM12878, K562, HepG2, SK-N-SH and A549 cells, respectively.

Assessing DNA chromatin accessibility and gene expression with Enformer. We replaced the endogenous sequence of putative regulatory regions with DNA-Diffusion or other endogenous sequences and considered Enformer predictions using a window of 393,216 base pairs around the region used for replacement. The resulting Enformer[43] predictions offered an approximate 128-bp resolution window for DNase I and CAGE tracks across all three cell lines. We averaged the extracted predictions across enhancer and promoter regions to generate a single value representing the predicted signal at each genomic coordinate.

Selecting random genomic regions and promoter regions. Random sequences were obtained using the BedTools random function, which selects random genomic coordinates of 200 bp. Promoter regions of protein-coding genes were extracted from GENCODE v42[66] and extended ±500 bp from the TSS to generate same-length intervals.

Normalization and scaling of the data

Because Enformer (DNase I and CAGE), ChromBPNet (DNase I) and the MPRA-predictor were trained using data with different sequencing depths, their baseline predictions exhibited a wide dynamic range, making it difficult to directly compare the difference in raw prediction scores across cell lines. Using quantile normalization, we adjusted these predictions and rescaled the data to a common range. This process helped to minimize the differences and biases, enabling more accurate comparisons and interpretations of the data across the loci and cell types.

For ChromBPNet predictions in all loci, we normalized the three cell lines' accessibility predictions, generating a similarly distributed set across K562, HepG2 and GM12878. Unlike ChromBPNet, which uses different models, Enformer outputs a separate prediction signal track for each cell type and data modality. Ultimately, all predictions were normalized together using the following groups: DNA-Diffusion, Training, Validation, Test, Promoters, Random Genome Regions, and the positive and negative control sequences previously validated on the basis of MPRA assays (see 'Selection of sequences for normalization and experimental validation' section). Signal intensity and specificity scores obtained from all oracles were normalized between 0 and 1 (Supplementary Methods 5).

Episomal experimental validation of DNA-Diffusion sequences with STARR-seq

Selection of sequences for normalization and experimental validation. To create a set of sequences that captured a broad range of potential outcomes in STARR-seq experiments, we generated a non-biased reference set by randomly sampling 400 sequences from the DNA-Diffusion model using a guidance scale of 1.0, without applying any preselection criteria. We also selected 400 endogenous DHS elements at random from the 47,873 sequences obtained from the DHS index dataset to serve as a baseline control.

For K562 positive sequences, 1,365 sequences were selected from the Peak Identifier for Nascent Transcript Starts (PINTS) database[67] based on prior validation by CRISPR, MPRA or STARR-seq assays. For HepG2-positive sequences, 112 sequences were similarly chosen using HepG2 sequences from the PINTS database. For GM12878-positive sequences, an initial set of 12,946 sequences identified by High-resolution Dissection of Regulatory Activity (HiDRA)[68] were intersected with the ENCODE DHS index to retain only regions with evidence of chromatin accessibility. The resulting DHS-overlapping sequences were ranked by HiDRA activity scores, and the top 4,000 were selected as high confidence. To define the final set of sequences for STARR-seq validation in each cell type, we averaged the predicted regulatory scores from Enformer (DNase I and CAGE models) and ChromBPNet (DNase I model) across the positive sequences selected in the previous step. The top 100 sequences with the highest averaged predicted activity were retained for experimental validation.

To select a set of negative sequences, we curated a negative control dataset starting with 7,000 sequences from the PINTS database. All sequences were previously functionally tested via MPRA or STARR-seq[27,28], although definitions of inactivity varied across source studies. These sequences were designated putative nonenhancers as they were prefiltered to exclude overlaps with ENCODE enhancer candidate *cis*-regulatory elements. While predominantly derived from K562 cell line experiments, this initial set was treated as cell-type agnostic. To generate cell-type-specific negative controls, we used ChromBPNet and Enformer computational models to predict regulatory activity for each sequence within the three target cell types. For each cell type, the 50 sequences exhibiting the lowest predicted activity scores were selected as the final negative control set.

To expand the number of DNA-Diffusion sequences evaluated in our library, we selected a total of 1,000 sequences from a previously generated pool of 300,000 sequences (100,000 per cell type: K562, GM12878 and HepG2), all created using a CFG guidance scale of 1.0. From this pool, we prioritized sequences that spanned a broad range of predicted DNase I accessibility intensities and cell-type specificities, as inferred from Enformer predictions. This strategy ensured that the selected sequences represented a diverse spectrum of chromatin accessibility profiles across the three cell types.

For more details about the experimental protocol and post-processing of experimental results, see Supplementary Methods 6.

In vitro validation of DNA-Diffusion and control sequences within the *AXIN2* locus

Selecting candidate sequences for downstream experimental validation. We evaluated a total of 105 sequences using the *AXIN2* endogenous replacement experiment (Supplementary Methods 7.1–7.4) using the following sequence groups: GM12878 DNA-Diffusion, GM12878 Positive, Shuffled GM12878, Negative Controls, K562 DNA-Diffusion and HepG2 DNA-Diffusion (Supplementary Table 7e). We also included the alternative and reference enhancer allele sequences as described below.

The sequences were inserted into a region situated 2.5 kb upstream of the *AXIN2* TSS. The putative chromatin accessibility or CAGE changes induced by the introduction of these sequences at the *AXIN2* locus was assessed using the Enformer model. Cell-type-specificity values were normalized following the procedures described previously (see 'Normalization and scaling of the data' section).

GM12878 DNA-Diffusion. We utilized the DNA-Diffusion model to sample 10,000 GM12878 cell type-specific sequences. These sequences were analyzed using Enformer to evaluate their accessibility by predicting the DNase I values at the replacement site upstream of the *AXIN2* TSS. From an initial pool of 10,000 candidate sequences, 60 sequences were selected across a range of predicted signal intensity and cell-type-specificity values. The selection strategy was designed to include a diverse set of sequences, enabling investigation of how different values for accessibility and *AXIN2* expression translate to experimentally observed effects on *AXIN2* expression.

GM12878 positive and negative controls. A set of positive control GM12878 sequences, previously validated experimentally using HiDRA, was obtained and filtered to keep only regions intersecting with GM12878 ENCODE DHS peaks. From this subset, ten sequences

with high predicted accessibility were selected for the final positive control group.

The negative control set of GM12878 sequences was selected from 7,097 sequences previously classified as nonenhancers in the PINTS database. These sequences exhibit minimal or no regulatory activity in MPRA and STARR-seq experiments and did not overlap genomic coordinates of candidate *cis*-regulatory elements (cCRE v3 hg38) from the SCREEN database[69]. The predicted DNase I activity at the replacement regions was determined for these sequences. A subset of five sequences was selected based on having the lowest values for DNase I signal intensity and DNase I specificity. In addition, we also included five shuffled (single nucleotide shuffle) sequences by shuffling previously selected DNA-Diffusion GM12878 sequences with the highest accessibility and *AXIN2* promoter reactivation predicted values.

K562 and HepG2 DNA-Diffusion. We generated 1,000 K562-specific and 1000 HepG2-specific DNA-Diffusion sequences. Enformer was used to predict the accessibility for these sequences.

For each respective cell type's 1,000 sequence set, a subset of five sequences was selected on the basis of their predicted high performance (high signal intensity and cell type-specificity) in K562 or HepG2 cells, while exhibiting minimal activity in GM12878 cells. The selection process involved ranking the sequences according to their Enformer predictions, considering the DNase I signal at the *AXIN2* TSS upstream replacement region. The signal intensity and specificity values were evaluated, and sequences with the highest values for DNase I signal and specificity were chosen. The specificity constraint was crucial for selecting sequences that displayed activation restricted to K562 or HepG2 cells, with minimal evidence of accessibility and *AXIN2* promoter reactivation in GM12878 cells.

Enformer predictions and visualization of the *AXIN2* locus sequences. To simulate the endogenous experiment, a custom version of the hg38 genome was generated for in silico prediction. This modified genome incorporated the insertion of several elements: LoxP sites, random barcodes, a fixed site, the endogenous untranslated region (UTR), the endogenous sequence, a 200-bp insertion region, a second fixed site and Lox2272 (Supplementary Table 7a–d). The integration of these components into the genome enabled a more accurate representation of the experimental conditions when performing in silico predictions.

Visualization of the *AXIN2* locus using nonprotective and protective alleles. The epigenetic tracks (assay for transposase-accessible chromatin using sequencing (ATAC-seq), H3K27ac and H3K4me3) from two individuals, one homozygous for the reference allele and the other homozygous for the alternative allele (rs143348853), were compared. Genetic variant information for these individuals was obtained from the 1000 Genomes Project[70], along with the corresponding epigenetic data from the European Nucleotide Archive-ENA (ATAC-seq) and ArrayExpress[71] (H3K4me3 and H3K27ac). BAM files were converted to the hg38 genome version using CrossMap v0.7.0[72] and transformed into BigWig tracks using DeepTools v3.5.4[73] with the following parameters: --binSize 10, --normalizeUsing RPKM and --extendReads.

TF enrichment analysis. To identify TF motifs associated with gene expression changes, the generated sequences were scanned using MOODS. The resulting motif hits were ranked in descending order based on their corresponding $\log_2$FC values. The prerank function from GSEApy (v1.1.3)[74] was then used to correlate the presence of each TF motif with its associated $\log_2$FC value with the following parameters: permutation_num=500, min_size=1 and max_size=5000. This analysis enabled the systematic assessment of the relationship between the occurrence of specific TF motifs and the observed changes in gene expression.

Quantification of endogenous *AXIN2* expression. For each of four replicates, genomic DNA and mRNA counts were quantified by sequencing the 12-nt barcodes corresponding to the designed DNA-Diffusion synthetic sequences. The DNA replicates were merged by barcode; replicates with no reads for a particular barcode had the count set to 0. The RNA replicates were merged following the same procedure. DNA barcode counts were then merged with the corresponding RNA counts by barcode, and barcodes sequenced only in mRNA reads (that is, at least one read across four replicates versus no reads across five DNA replicates) were omitted from further analyses.

The full listing of paired 12-nt barcodes with designed DNA-Diffusion synthetic sequences was filtered to include only barcodes with a minimum of three reads in the mapping step. The DNA and RNA barcodes were then assigned to their corresponding synthetic sequences using this mapping, and barcodes with no assigned synthetic sequence were omitted. For each barcode, two key statistics were calculated: the $\log_2$ ratio of the sum of RNA counts to the sum of DNA counts, and the difference between this ratio and the median ratio across all barcodes for the corresponding synthetic sequence ('ratio_diff'). Outlier barcodes were removed if ratio_diff was greater than five times the median absolute deviation of ratio_diff across all barcodes within the same decile of total RNA counts. After filtering per replicate, any RNA count corresponding to a DNA count of 0 was set to 0. The transcriptional activity was analyzed using the 'mpra' package[75], following the same methodology mentioned in the STARR-seq session (Supplementary Table 7e).

Training and benchmarking computational regulatory element generators

CODA/Malinois[17], DeepMEL2 (ref. 18) and a WGAN[18] were trained using the original DNA-Diffusion dataset with adaptations to the training. For a detailed explanation of each model-specific training pipeline, see Supplementary Methods 8.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

To train and test the DNA-Diffusion model, we used a curated set of DHSs from Meuleman et al.[32]. A processed version containing the cell-type-specific sequences used to train and test our DNA-Diffusion model is available via GitHub at https://raw.githubusercontent.com/pinellolab/DNA-Diffusion/main/data/K562_hESCT0_HepG2_GM12878_12k_sequences_per_group.txt, along with notebooks containing the exact procedures used to preprocess the sequences. All the supplementary data in support of this work (Supplementary Data 1–4) are available via Zenodo at https://doi.org/10.5281/zenodo.17420419 (ref. 76). The experimental STARR-seq mRNA and DNA sequence data with the different replicates can be accessed via GEO at GSE293971. The sequencing data in support of the *AXIN2* experimental validation can be accessed using the gRNA (GSE269915), mRNA (GSE269912) and Nanopore (GSE269921) GEO accession numbers. The BAM/CRAM files used to process and visualize the individual reference and indel (rs143348853) ATAC-seq data were collected from the European Nucleotide Archive-ENA: Reference (HG00233) and Indel (HG00247). The histone marks (H3K4me3 and H3K27ac) utilized to visualize these individuals were collected from the functional genomics data collection (ArrayExpress) using the ID NA12282 for the reference individual and NA11931 ID for the indel individual. The model weights for DNA-Diffusion, MPRA predictor, DeepMEL, CODA and WGAN (one per cell type) (Supplementary Data 5–9) are available via Zenodo at https://doi.org/10.5281/zenodo.17420419 (ref. 76). The 'mpra/voom' scripts used to process EXTRA-seq and STARR-seq data by analyzing DNA and mRNA counts and computing the mRNA/DNA logFC across

experimental replicates (Supplementary Data 10) are also available via Zenodo at https://doi.org/10.5281/zenodo.17420419 (ref. 76). Source data are provided with this paper.

Code availability

The source code for DNA-Diffusion is publicly available via GitHub at https://github.com/pinellolab/DNA-Diffusion (release version 0.1.1) and is also available via Zenodo at https://doi.org/10.5281/zenodo.17420419 (ref. 76). The repository includes trained model weights, dependency requirements and working examples demonstrating the complete workflow from sequence generation to evaluation. Code and scripts used to perform all experiments described in this Article, including benchmarking analyses, are available via Zenodo at https://doi.org/10.5281/zenodo.17420419 (ref. 76), along with model weights for WGAN, CODA and DeepMEL.

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Acknowledgements

We acknowledge B. Wang, V. Subasri, M. Consens, R. Ma, J. Lin, Z. Li, P. Cornejo, N. Zanichelli, members of the OpenBioML.org community, and the Pinello laboratory for helpful feedback and stimulating discussions. We thank Z. Li and Y.-H. Yang for research assistance. We are grateful to A. Kundaje and A. Pampari for their generous support in using the ChromBPnet models. We thank StabilityAI for providing access to their high-performance computing cluster, which enabled the training and testing of our models. L.P. was partially supported by 1R35HG010717-01 and the Rappaport MGH Research Scholar Award 2024-2029, as well as a Krantz Center Spark Award. W.M. was partially supported by 1R35HG011317.

Author contributions

L.F.D. W.M. and L.P. conceptualized the project and its scope. L.F.D., P.B. and S.S. performed the preprocessing of the DHS data utilized to train the diffusion model. L.F.D., Z.N. and S.S. designed, implemented and trained the diffusion model. Z.M.P., L.F.D. and S.S. performed the model interpretation and TF motif analyses. A.J.R., S.S. and T.A.K. conducted computational MPRA training and analysis. S.G., L.P., L.F.D. and S.S. performed the ChromBPNet analysis. C.M.V.C., L.F.D., S.S. and Z.M.P. conducted the BLAT analysis. A.W., T.M.T., L.F.D. and S.S. assisted with Enformer analyses. M.M, N.W., C.M.V.C., W.C., C.S. and M.B. assisted with model testing and exploration. L.F.D. curated the control datasets. L.K.L. and E.S.W. designed and performed the STARR-seq experiment. D.T.H., L.F.D., M.I.L., J.D.R., S.S. and E.S.W. performed postprocessing and downstream analysis of the STARR-seq results. J.F.K.-S. performed the endogenous replacement experiment. J.F.K.-S., J.D.R., G.-H.H. and V.G. performed the postprocessing of the experimental data. L.F.D., S.S., J.F.K.-S., Z.M.P., A.J.R., S.G., J.D.R., Z.N., A.W., N.W., T.M.T., M.M., V.G., E.W., B.D., W.M. and L.P. assisted with manuscript writing. L.F.D., S.S., Z.M.P. and L.P. prepared the manuscript and associated figures with input from all authors. We note that J.F.K.-S., Z.M.P. and L.K.L. contributed equally to this work as co-second authors. L.P. supervised the project.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41588-025-02441-6.

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Peer review information *Nature Genetics* thanks the anonymous reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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Last updated by author(s): Oct 24, 2025

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data was collected from public repositories details are provided in the data section.
Data analysis	The source code of DNA-Diffusion is publicly available at https://github.com/pinellolab/DNA-Diffusion. Processed data, models weights, and custom scripts used to perform analyses were uploaded to Zenodo https://zenodo.org/records/151829699 We also provide below a list of the tools used in the manuscript:. Enformer (https://github.com/google-deepmind/deepmind-research/tree/f5de0ede8430809180254ee957abf36ed62579ef/enformer) ChromBPNet v0.1.7 Bedtools v2.30.0 Python v3.10 accelerate v0.24.1 click v8.1.7 einops v0.7.0 genomepy v0.16.1 gimmemotifs v0.18.0 gtfparse v1.3.0 jupyterlab v4.0.9 matplotlib v3.8.1

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To train and test the DNA-Diffusion model, we used a curated set of DHSs from Meuleman et al. A processed version containing the cell type-specific sequences used to train and test our DNA-Diffusion model can be found in the project GitHub (https://raw.githubusercontent.com/pinellolab/DNA-Diffusion/main/data/K562_hESCT0_HepG2_GM12878_12k_sequences_per_group.txt), along with notebooks containing the exact procedures used to pre-process the sequences. All the supplementary data in support of this work (Supplementary Data 1-4) have been uploaded to Zenodo DOI: 10.5281/zenodo.17420419. The experimental STARR-Seq mRNA and DNA sequence data with the different replicates can be accessed in the GEO (GSE293971). The sequencing data in support of the AXIN2 experimental validation can be accessed using the gRNA (GSE269915), mRNA (GSE269912), and Nanopore (GSE269921) GEO accession numbers. The BAM/CRAM files used to process and visualize the individual reference and indel (rs143348853) ATAC-Seq data were collected from the European Nucleotide Archive-ENA: Reference (HG00233) and Indel (HG00247). The histone marks (H3K4me3 and H3K27ac) utilized to visualize these individuals were collected from the functional genomics data collection (ArrayExpress) using the ID NA12282 for the reference individual and NA11931 ID for the indel individual.

The model weights for DNA-Diffusion, MPRA predictor, DeepMEL, CODA, and WGAN (one per cell type) (Supplementary Data 5-9) are available on Zenodo (DOI: 10.5281/zenodo.17420419.). The “mpra/voom” scripts used to process EXTRA-Seq and STARR-Seq data by analyzing DNA and mRNA counts and computing the mRNA/DNA log fold change across experimental replicates (Supplementary Data 10) are available on Zenodo (DOI: 10.5281/zenodo.17420419).

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Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used.* *Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected.* *Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
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Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The number of sequences used to train each cell type was 12k and it was selected to balance the training sequences between cell types. Since GM12878 cells only have 12k cell-specific peaks following the criteria described in our methods section this was the selected number of sequences. Even using this amount of cells our results showed a capacity to retain unique properties of each cell-type and generate a diverse set of new sequences.
Data exclusions	Data exclusion was only executed when balancing the number of DHS training sequences for each cell type. These criteria was described in the methods and tried to prioritize peaks that presented concordance and multiple DNAse experiments and with the lower presence in other Encode cell types
Replication	The Genomic DNA (gDNA) and mRNA were extracted from a total of five million cells in four technical replicates using the mini AllPrep DNA/RNA extraction kit. This number of replicates has been demonstrated to be enough to capture the differential expression of genes in thousands of experiments across the literature.
Randomization	We used bedtools random command to select a subset of 200bp genomic sequences using from random human coordinates
Blinding	Blinding was not applicable for our experimental design

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study	
☒	☐	Antibodies
☐	☒	Eukaryotic cell lines
☒	☐	Palaeontology and archaeology
☒	☐	Animals and other organisms
☒	☐	Clinical data
☒	☐	Dual use research of concern
☒	☐	Plants

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	MEC1, an EBV positive B-cell line) was created with a stably integrated CRE recombinase, as well as a CRE-recombinase mediated cassette-exchange (RMCE) landing pad (heterozygous) including a loxP and a lox2722 site flanking a CMV-GFP construct (acc 497, DSMZ) -GM12878 (female CEPH/UTAH pedigree, B-lymphocyte derived) cells were purchased from Coriell Institute for Medical Research -HepG2 cells (male, hepatocellular carcinoma derived) gifted from N.Turner (VCCRI) -K562 line was gifted from S.Dunwoodie (VCCRI).
Authentication	Authentication by STR analysis according to the global standard ANSI/ATCC ASN-0002.1-2021 (2021) resulted in an authentic STR profile of the reference STR database, identical to sister cell line MEC-2 (ACC 500) Leibniz Institute - DSMZ
Mycoplasma contamination	Cells were confirmed negative for mycoplasma.
Commonly misidentified lines (See ICLAC register)	Not misidentification found in the ICLAC v 13

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*